

Ryun-Han Koo

其他

基本信息

姓名: Ryun-Han Koo

所属院校: 首尔国立大学

地区: South Korea

学科领域

微电子科学与工程

材料物理

集成电路设计与集成系统

研究领域

薄膜材料与表面物理

微电子器件与制造

纳米电子与先进工艺

存储器设计与架构

半导体与光电子材料

材料物理与化学

低功耗电路与能量管理

薄膜与表面技术

个人简介

Ryun-Han Koo is a distinguished academic affiliated with Seoul National University. His research focuses on semiconductor devices, ferroelectric devices, and neuromorphic technology. Koo has made significant contributions to the field through his publications in high-impact journals such as Advanced Science, Sensors and Actuators B: Chemical, and IEEE Electron Device Letters. His work often explores the integration of ferroelectric materials in transistor-based devices and their applications in neuromorphic systems and gas sensors.

工作经历

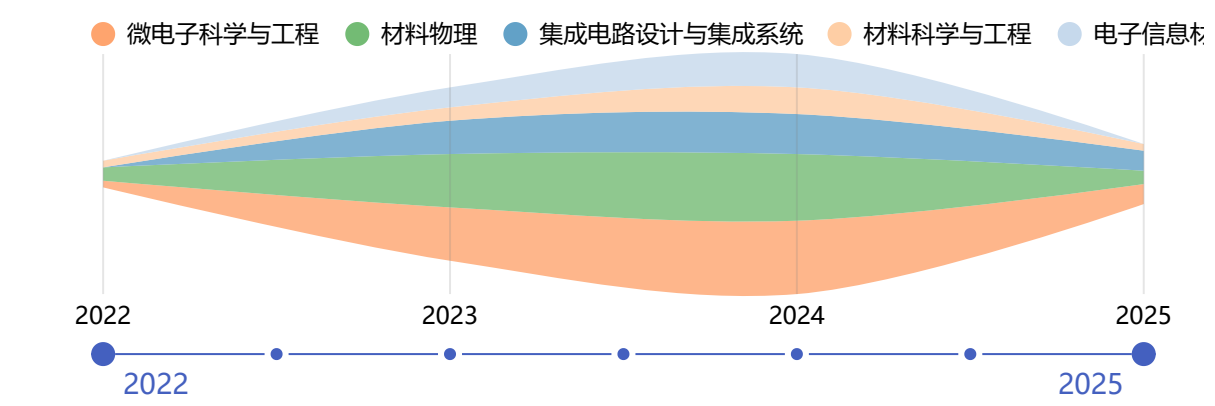
Seoul National University

教育经历

Seoul National University - null

null - null

近年发表的论文



Does a large response suffice?: Thermally stable and low noise Si-doped IZO thin-film transistor-type gas sensors(vol 422 , 136498 , 2025)
SENSORS AND ACTUATORS B-CHEMICAL
Shin, Wonjun; Lee, Sunjin; Koo, Ryun-Han; Kim, Jangsaeng; Lee, Sang Yeol; Lee, Sung-Tae
被引用次数: 0 2025

Vertical NAND flash memory-based matrix-matrix multiplication scheme for energy-efficient attention score computation
NEUROCOMPUTING
Ko, Jonghyun; Park, Sung-Ho; Im, Jiseong; Kim, Jeonghyun; Koo, Ryun-Han; Yang, Yeongheon; Lee, Jong-Ho
被引用次数: 0 2025

Does a large response suffice?: Thermally stable and low noise Si-doped IZO thin-film transistor-type gas sensors
SENSORS AND ACTUATORS B-CHEMICAL
Shin, Wonjun; Lee, Sunjin; Koo, Ryun-Han; Kim, Jangsaeng; Lee, Sang Yeol; Lee, Sung-Tae
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Stochastic behavior of random telegraph noise in ferroelectric devices: Impact of downscaling and mitigation strategies for neuromorphic applications
CHAOS SOLITONS & FRACTALS
Koo, Ryun-Han; Shin, Wonjun; Lee, Sung-Tae; Kwon, Daewoong; Lee, Jong-Ho
被引用次数: 0 2025

Retention Improvement in Vertical NAND Flash Memory Using Electron Back-Tunneling
IEEE ELECTRON DEVICE LETTERS
Park, Sung-Ho; Yoo, Ho-Nam; Back, Jong-Won; Cho, Joonhyung; Yang, Yeongheon; Im, Jiseong; Koo, Ryun-Han; Ko, Jonghyun; Lee, Jong-Ho
被引用次数: 0 2025

NOR-Type Flash Array Based on Four-Terminal TFT Synaptic Devices Capable of Selective Program/Erase Exploiting Fowler-Nordheim Tunneling

IEEE ELECTRON DEVICE LETTERS

Hwang, Joon; Park, Min-Kyu; Kang, Won-Mook; Koo, Ryun-Han; Lee, Kyu-Ho; Kwon, Dongseok; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 0

2024

Vertical AND-Type Flash TFT Array Capable of Accurate Vector-Matrix Multiplication Operations for Hardware Neural Networks

IEEE ELECTRON DEVICE LETTERS

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被引用次数: 1

2024

Robust 1/f Noise Unaffected by Program/Erase Cycling-Induced Damage in Ferroelectric Schottky Barrier FETs

IEEE ELECTRON DEVICE LETTERS

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2024

On-Chip Learning in Vertical NAND Flash Memory Using Forward-Forward Algorithm

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Park, Sung-Ho; Ko, Jonghyun; Lee, In-Seok; Koo, Ryun-Han; Kim, Jeong-Hyun; Yang, Yeongheon; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 1

2024

Proposition of optimal self-curing method in horizontal-floating gate FET-type gas sensors for reliability improvement

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Lee, Sung-Tae; Lee, Jong-Ho

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2024

Chemisorption Manipulation by Adjusting the Carrier Concentration of the Adsorbent and Its Application to Adsorbate Identification

ACS MATERIALS LETTERS

Jung, Gyuweon; Choi, Kangwook; Ju, Suyeon; Kim, Jaehyeon; Shin, Wonjun; Lee, Seongi; Yeom, Gyuho; Koo, Ryun-Han; Joo, Young-Chang; Choi, Woo Young; Han, Seungwu; Lee, Jong-Ho

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2024

Strain-Stress Impact on Ferroelectric Devices: A Multilayer Analysis and Optimization Strategy for Neural Networks

ACS MATERIALS LETTERS

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2024

Examination of Ferroelectric Domain Dynamics in HZO Under Endurance Cycling Stress

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Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Kim, Seungwhan; Jung, Gyuweon; Lee, Sung-Tae; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2024

Analog reservoir computing via ferroelectric mixed phase boundary transistors

NATURE COMMUNICATIONS

Kim, Jangsaeng; Park, Eun Chan; Shin, Wonjun; Koo, Ryun-Han; Han, Chang-Hyeon; Kang, He Young; Yang, Tae Gyu; Goh, Youngin; Lee, Kilho; Ha, Daewon; Cheema, Suraj S.; Jeong, Jae Kyeong; Kwon, Daewoong

被引用次数: 1

2024

SNNSim: Investigation and Optimization of Large-Scale Analog Spiking Neural Networks Based on Flash Memory Devices

ADVANCED INTELLIGENT SYSTEMS

Ko, Jong Hyun; Kwon, Dongseok; Hwang, Joon; Lee, Kyu-Ho; Oh, Seongbin; Kim, Jeonghyun; Im, Jiseong; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong-Ho

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2024

Multifunctional In-Memory Analog-to-Digital Converter for Next-Gen Compute-in-Memory Systems

ADVANCED INTELLIGENT SYSTEMS

Im, Jiseong; Ko, Jonghyun; Hwang, Joon; Kim, Jangsaeng; Shin, Wonjun; Koo, Ryun-Han; Park, Minkyu; Park, Sung-Ho; Choi, Woo Young; Kim, Jae-Joon; Lee, Jong-Ho

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2024

Unveiled Influence of Sub-gap Density of States on Low-Frequency Noise in Si-Doped ZnSnO TFTs: Does Correlated Mobility Fluctuation Model Suffice?

ADVANCED ELECTRONIC MATERIALS

Shin, Wonjun; Lee, Ji Ye; Koo, Ryun-Han; Kim, Jangsaeng; Lee, Jong-Ho; Lee, Sang Yeol; Lee, Sung-Tae

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2024

Effects of Charge Imbalance on Field-Induced Instability of HfO₂-Based Ferroelectric Tunnel Junctions

ADVANCED ELECTRONIC MATERIALS

Shin, Wonjun; Han, Chang-Hyeon; Kim, Jangsaeng; Koo, Ryun-Han; Min, Kyung Kyu; Kwon, Daewoong

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2024

Toward Understanding Temperature and Bias Instabilities of Anti-ambipolar Transistors via Low-Frequency Noise Spectroscopy

SMALL

Song, Jaechan; Han, Youngmin; Koo, Ryun-Han; Seo, Junghye; Yoo, Hocheon; Shin, Wonjun

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2024

Proposition of Adaptive Read Bias: A Solution to Overcome Power and Scaling Limitations in Ferroelectric-Based Neuromorphic System

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Seungwhan; Im, Jiseong; Park, Sung-Ho; Ko, Jong Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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2024

Toward Ideal Low-Frequency Noise in Monolayer CVD MoS₂ FETs: Influence of van der Waals Junctions and Sulfur Vacancy Management

ADVANCED SCIENCE

Shin, Wonjun; Byeon, Junsung; Koo, Ryun-Han; Lim, Jungmoon; Kang, Jung Hyeon; Jang, A-Rang; Lee, Jong-Ho; Kim, Jae-Joon; Cha, Seungnam; Pak, Sangyeon; Lee, Sung-Tae

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2024

Polarization Pruning: Reliability Enhancement of Hafnia-Based Ferroelectric Devices for Memory and Neuromorphic Computing

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Jangsaeng; Yim, Jiyong; Ko, Jonghyun; Jung, Gyuweon; Im, Jiseong; Park, Sung-Ho; Kim, Jae-Joon; Cheema, Suraj S.; Kwon, Daewoong; Lee, Jong-Ho

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2024

All-Ferroelectric Spiking Neural Networks via Morphotropic Phase Boundary Neurons

ADVANCED SCIENCE

Kim, Jangsaeng; Park, Eun Chan; Shin, Wonjun; Koo, Ryun-Han; Im, Jiseong; Han, Chang-Hyeon; Lee, Jong-Ho; Kwon, Daewoong

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2024

A New Back-End-Of-Line Ferroelectric Field-Effect Transistor Platform via Laser Processing

SMALL

Kim, Sang Woo; Shin, Wonjun; Koo, Ryun-Han; Kim, Jangsaeng; Im, Jiseong; Koh, Dooyong; Lee, Jong-Ho; Cheema, Suraj S.; Kwon, Daewoong

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2024

Charge Transport Advancement in Anti-Ambipolar Transistors: Spatially Separating Layer Sandwiched between N-Type Metal Oxides and P-Type Small Molecules

ADVANCED FUNCTIONAL MATERIALS

Han, Youngmin; Lee, Subin; Kim, Minseo; Shin, Wonjun; Lee, Ho kyung; Koo, Ryun-Han; Lee, Sung-Tae; Kim, Chang-Hyun; Yoo, Hocheon

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2024

Stochasticity in ferroelectric memory devices with different bottom electrode crystallinity

CHAOS SOLITONS & FRACTALS

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2024

Low-frequency noise of MoTe₂ transistor: effects on ambipolar carrier transport and CYTOP doping

DISCOVER NANO

Shin, Wonjun; Lee, Dong Hyun; Ko, Raksan; Koo, Ryun-Han; Yoo, Hocheon; Lee, Sung-Tae

Unveiling Resistance Switching Mechanisms in Undoped HfO_x Ferroelectric Tunnel Junction Using Low-Frequency Noise Spectroscopy

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Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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Annealing Ambient and Film Thickness Dependent NO₂ Response and 1/f Noise Characteristics of IGZO Resistor-Type Gas Sensors

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Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Lee, Sung-Tae; Lee, Jong-Ho

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Effect of Carrier Transport Process on Tunneling Electroresistance in Ferroelectric Tunnel Junction

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2023

Recovery of off-state stress-induced damage in FET-type gas sensor using self-curing method

DISCOVER NANO

Shin, Wonjun; Jeong, Yujeong; Kim, Mingyu; Lee, Jungsoo; Koo, Ryun-Han; Hong, Seongbin; Jung, Gyuweon; Kim, Jae-Joon; Lee, Jong-Ho

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Effects of Sensor Platform Scaling on Signal-to-Noise Ratio in the Resistor-and Horizontal Floating-Gate FET-Type Gas Sensors

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Hardware-Based Ternary Neural Network Using AND-Type Poly-Si TFT Array and Its Optimization Guideline

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kwon, Dongseok; Park, Min-Kyu; Kang, Won-Mook; Hwang, Joon; Koo, Ryun-Han; Bae, Jong-Ho; Lee, Jong-Ho

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2023

Cointegration of the TFT-Type AND Flash Synaptic Array and CMOS Circuits for a Hardware-Based Neural Network

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2023

Voltage scheme for string-select transistors to improve inhibition characteristics during 1-bit erase in vertical NAND flash

APPLIED PHYSICS LETTERS

Park, Sung-Ho; Yoo, Ho-Nam; Yang, Yeongheon; Back, Jong-Won; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho

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2023

1/f Noise in Synaptic Ferroelectric Tunnel Junction: Impact on Convolutional Neural Network

ADVANCED INTELLIGENT SYSTEMS

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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2023

Low-frequency noise characteristics of indium-gallium-zinc oxide ferroelectric thin-film transistors with metal-ferroelectric-metal-insulator-semiconductor structure

APPLIED PHYSICS LETTERS

Shin, Wonjun; Park, Eun Chan; Koo, Ryun-Han; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 3

2023

Effects of temperature and DC cycling stress on resistive switching mechanisms in hafnia-based ferroelectric tunnel junction

APPLIED PHYSICS LETTERS

Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwak, Been; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

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2023

Self-Curable Synaptic Ferroelectric FET Arrays for Neuromorphic Convolutional Neural Network

ADVANCED SCIENCE

Shin, Wonjun; Im, Jiyong; Koo, Ryun-Han; Kim, Jaehyeon; Kwon, Ki-Ryun; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho; Kwon, Daewoong

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2023

In-Memory-Computed Low-Frequency Noise Spectroscopy for Selective Gas Detection Using a Reducible Metal Oxide

ADVANCED SCIENCE

Shin, Wonjun; Kim, Jaehyeon; Jung, Gyuweon; Ju, Suyeon; Park, Sung-Ho; Jeong, Yujeong; Hong, Seongbin; Koo, Ryun-Han; Yang, Yeongheon; Kim, Jae-Joon; Han, Seungwu; Lee, Jong-Ho

被引用次数: 6

2023

Analog Synaptic Devices Based on IGZO Thin-Film Transistors with a Metal-Ferroelectric-Metal-Insulator-Semiconductor Structure for High-Performance Neuromorphic Systems

ADVANCED INTELLIGENT SYSTEMS

Kwon, Dongseok; Park, Eun Chan; Shin, Wonjun; Koo, Ryun-Han; Hwang, Joon; Bae, Jong-Ho; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2023

Power and Area-Efficient XNOR-AND Hybrid Binary Neural Networks Using TFT-Type Synaptic Device

IEEE ELECTRON DEVICE LETTERS

Lee, In-Seok; Kim, Hyeongsu; Park, Min-Kyu; Hwang, Joon; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong-Ho

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2023

Low-frequency noise in gas sensors: A review

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Choi, Kangwook; Shin, Hunhee; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong -Ho

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2023

Comparative Analysis of n- and p-Type Ferroelectric Tunnel Junctions Through Understanding of Non-FE Resistance Switching

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Lee, Kyungmin; Park, Sung-Ho; Im, Jiseong; Ko, Jong-Hyun; Kim, Jeong-Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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Unraveling Threshold Voltage Instability in Ferroelectric Junctionless FETs Using Low-Frequency Noise Measurement With Base Bias

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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2023

Optimizing Post-Metal Annealing Temperature Considering Different Resistive Switching Mechanisms in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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Channel-Length-Dependent Low-Frequency Noise Characteristics of Ferroelectric Junctionless Poly-Si Thin-Film Transistors

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Kim, Sangwoo; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Deok-Hwang; Kwon, Daewoong; Lee, Jong-Ho

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2023

Investigation of Low-Frequency Noise Characteristics of Ferroelectric Tunnel Junction: From Conduction Mechanism and Scaling Perspectives

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kwon, Dongseok; Koo, Ryun-Han; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

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Synergistic improvement of sensing performance in ferroelectric transistor gas sensors using remnant polarization

MATERIALS HORIZONS

Shin, Wonjun; Yim, Jiyong; Bae, Jong-Ho; Lee, Jung-Kyu; Hong, Seongbin; Kim, Jaehyeon; Jeong, Yujeong; Kwon, Dongseok; Koo, Ryun-Han; Jung, Gyuweon; Han, Changhyeon; Kim, Jeonghan; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

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2022

Variability analysis of ferroelectric FETs in program operation using low-frequency noise spectroscopy

APPLIED PHYSICS LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

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Highly Efficient Self-Curing Method in MOSFET Using Parasitic Bipolar Junction Transistor

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2022

Frobenius numbers of Pythagorean triples

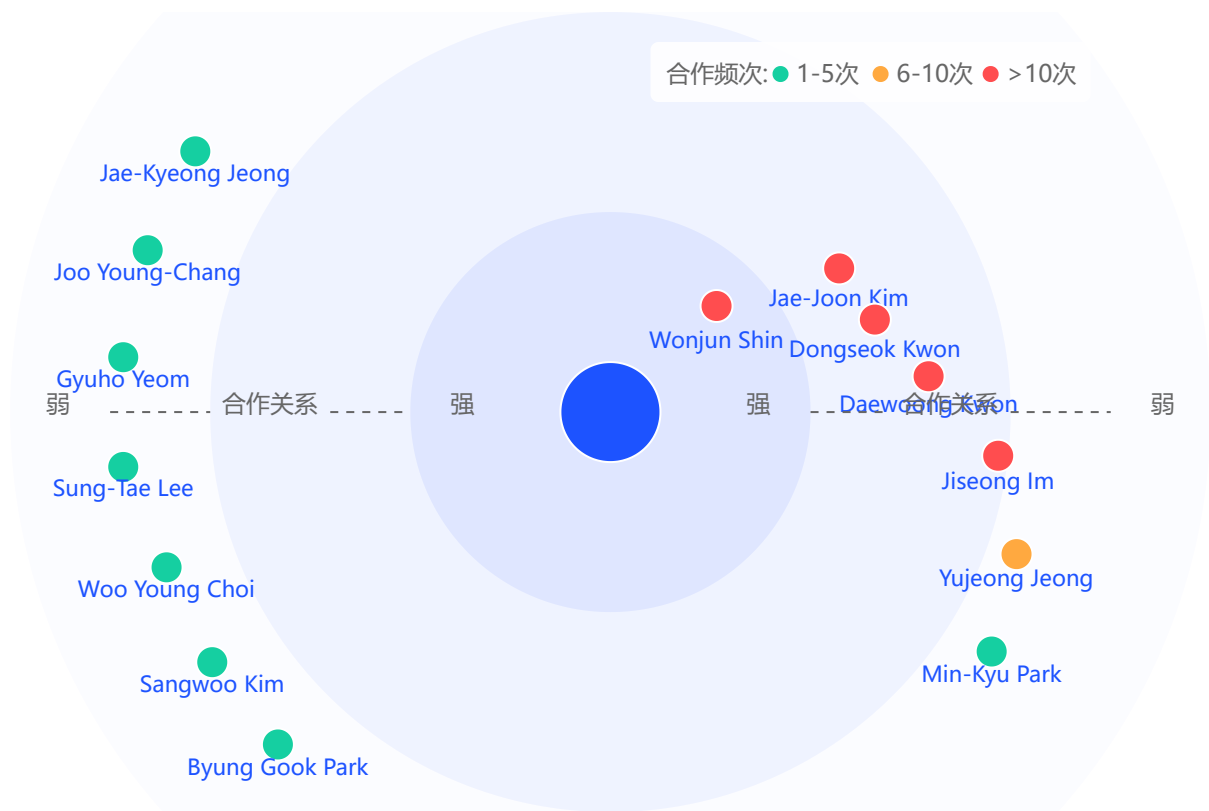
INTERNATIONAL JOURNAL OF NUMBER THEORY

Gil, Byung Keon; Han, Ji-Woo; Kim, Tae Hyun; Koo, Ryun Han; Lee, Bon Woo; Lee, Jaehoon; Nam, Kyeong Sik; Park, Hyeon Woo; Park, Poo-Sung

被引用次数: 7

2015

学术合作



Wonjun Shin

成均馆大学

合作次数：35

Jae-Joon Kim

首尔国立大学

合作次数：23

Dongseok Kwon

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